

SciText2Eq: Assessing LLMs for Explainable Equation Generation for Scientific Creativity

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Abstract

This work investigates the ability of large language models (LLMs) to generate mathematical equations from scientific texts. Prior work faces challenges in unstructured grounding, multi-equation dependency, and human-aligned evaluation. To this end, we construct a dataset of AI research papers, pairing contextual passages with ground-truth equations and variable descriptions. We develop an explainable equation generation workflow and evaluate it across diverse open- and closed-source LLM backbones. We introduce an evaluation protocol combining automatic metrics, LLM-based rubrics, and human judgments to assess accuracy, explainability, and human-LLM alignment. Results indicate that LLMs perform moderately on lexical- and syntactic-based similarity, while struggling with semantic accuracy. Comparisons between LLM-based evaluations and human judgments reveal limited alignment, highlighting challenges in using LLMs to assess equation quality. These findings offer insights for improving equation generation models and developing more reliable evaluation methods for scientific text. We provide code and data for reproducibility. ¹

1 Introduction

Mathematical equations underpin scientific knowledge representation and reasoning (Bais, 2025). Recent research include equation recognition (Zhou et al., 2022), equation retrieval (Wang et al., 2021; Vemuganti et al., 2025), mathematical language understanding (Peng et al., 2021; Scarlatos and Lan, 2023), equation reasoning (Meadows et al., 2025; Yu et al., 2025). As scientific literature grows denser, generating and explaining equations in natural language has become central to scientific creativity (Yasunaga and Lafferty, 2019).

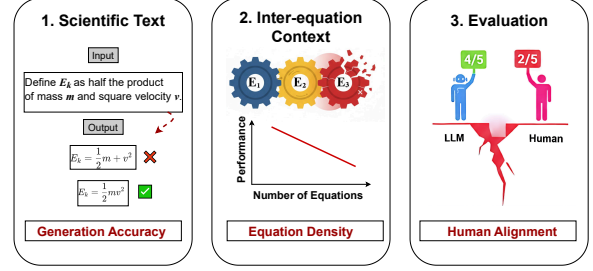


Figure 1: Three key challenges in generating equations from unstructured text.

A key line of related work is equation discovery, or symbolic regression (SR), which recovers mathematical expressions describing relationships among observed variables (Dong and Zhong, 2025). Recent approaches leverage LLMs to propose and refine candidate equations using programming (Shojaee et al., 2025a), reasoning (Wang et al., 2025), and benchmarking (Shojaee et al., 2025b). In contrast, equation generation focuses on producing expressions conditioned on textual context. Prior work includes extracting single equations from unstructured narrative text (Zhang et al., 2023; Liang et al., 2023; Zong and Krishnamachari, 2023). TopicEq jointly models topics and equations to align expressions with scientific text (Yasunaga and Lafferty, 2019), yet its dataset is not publicly available and contexts remain relatively simple.

In real scientific texts, equations appear in unstructured narratives, requiring explanation, inter-equation reasoning, and semantic fidelity. There are three major challenges: (1) accurately generating mathematical expressions from scientific texts where equations are interleaved with natural language descriptions; (2) handling multiple, non-independent equations within the same document and understanding how equation density and inter-equation context affect generation accuracy; (3) evaluating generated equations in a comprehensive way and aligning large language

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¹<https://github.com/YifanMo727/SciText2Eq>

model (LLM)-based judgments with human assessments, while maintaining explainability of the process. These challenges reveal a gap in generating publication-quality scientific equations (Figure 1).

In this work, we propose **SciText2Eq**, a comprehensive workflow for scientific discovery by LLMs in real-world scientific text. We construct a dedicated dataset of AI research papers, selecting sections rich in equations to provide realistic multi-equation contexts. Equations and their corresponding variable descriptions are sequentially generated by LLMs from the constructed dataset, based on all context and equations preceding the target equation within each paper. Generated equations are evaluated using a multi-level protocol that combines: (1) automatic metrics measuring lexical and syntactic similarity; (2) rubric-based LLM-as-judge evaluation assessing validity, coverage, clarity, appropriateness, and equivalence of both equations and their accompanying descriptions; and (3) human evaluation to quantify alignment between LLM and expert judgment.

The contributions of this work are threefold:

- A workflow for scientific equation generation using LLMs, providing both variable-level explanations and interpretable evaluation rationales.
- A comprehensive multi-level evaluation framework, combining automatic metrics, LLM-based rubric metrics, and human judgment, along with an extensible benchmark dataset.
- An empirical study of equation density, revealing its impact on equation generation performance.

2 Related work

2.1 Equation Discovery and Generation

Equation discovery, also known as SR, aims to recover mathematical expressions from observational data or variables. Traditional SR methods include evolutionary algorithms (Koza, 1994) and gradient-based optimization (Petersen et al., 2021), while more recently, LLMs have been applied to these tasks. As shown in Appendix A, most approaches use variable pairs as primary input, with general SR methods performing equation discovery without additional guidance (Merler et al., 2024), and domain-specific methods incorporating structured information such as scientific priors (Wang et al., 2025; Song et al., 2025), symbol libraries (Du et al., 2024), or brief natural language descriptions of tasks to constrain the search space and guide equation generation (Shojaee et al., 2025a,b).

In contrast, equation generation produces mathematical expressions from textual context rather than variable pairs. Early work uses natural language math word problems (MWP) from public datasets (Zhang et al., 2023; Liang et al., 2023; Zong and Krishnamachari, 2023), such as MathQA (Amini et al., 2019) and Math23K (Wang et al., 2017), which mainly contain synthetic arithmetic or algebra problems where text maps directly to equations, requiring little domain knowledge or reasoning. Some approaches also generate equations by modeling the semantic roles of numbers and operators within the problem context (Chiang and Chen, 2019). Recent work considers real scientific text by pairing each equation with surrounding sentences (Yasunaga and Lafferty, 2019), but it focuses on local context and does not fully capture dependencies across broader sections of a paper.

In summary, while existing work has advanced equation discovery and generation, most methods either focus on structured or synthetic inputs (Merler et al., 2024; Shojaee et al., 2025a) or assume that equations in real scientific texts depend primarily on local context (Yasunaga and Lafferty, 2019). This leaves a gap in generating equations from complex, long-form scientific narratives that require integrating global context, which is the focus of our work.

2.2 Equation Density

Most benchmarks for equation discovery focus on SR tasks with short, independent variable pairs, while domain-specific methods may use priors, symbol libraries, or brief task descriptions (Wang et al., 2025; Shojaee et al., 2025b; Du et al., 2024), yet equations remain isolated without broader context. Similarly, prior work on equation generation often considers independent problems or short text snippets, generating each equation separately (Zhang et al., 2023; Liang et al., 2023) or using only limited local context (Yasunaga and Lafferty, 2019). Real scientific papers, however, contain multiple interdependent equations within long narratives.

Overlooking broader context and inter-equation dependencies can limit a model’s ability to capture semantic structure. To better reflect generation performance, our dataset preserves full narrative context, providing all preceding text and equations for each target formula, which enables analysis of how equation density affects generation quality.

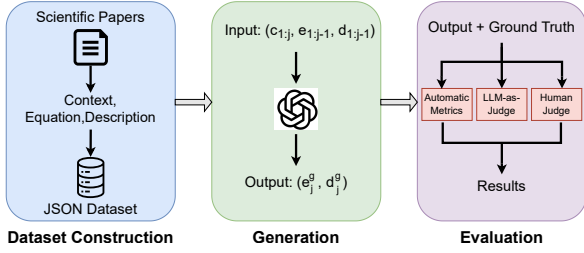


Figure 2: The proposed workflow for explainable equation generation in scientific discovery.

2.3 Evaluation and Human Alignment

Prior work on equation discovery and generation has largely relied on automated metrics. In symbolic regression, common metrics such as NMSE (Wang et al., 2025; Shojaei et al., 2025a), NRMSE (Du et al., 2024), R^2 (Merler et al., 2024), quantify how closely generated equations match observed data (Table 5). For equation generation from natural language, evaluation often focuses on numerical or structural equivalence. Models are assessed on whether generated equations yield the correct numeric results (Zhang et al., 2023; Liang et al., 2023) or on language-model-based metrics such as perplexity and syntax error rate for LaTeX sequences (Yasunaga and Lafferty, 2019). Some approaches incorporate LLMs as evaluators (Shojaei et al., 2025b), but they generally assess only one dimension at a time and seldom consider semantic consistency, contextual appropriateness, or clarity of variable roles.

Human evaluation has been relatively limited in prior work. In equation discovery, few studies compare LLM-generated outputs with human judgment (Shojaei et al., 2025b), and in equation generation, some works evaluate numeric equivalence or label matching against human annotations (Zong and Krishnamachari, 2023). Comprehensive rubric-based human evaluation remains insufficient, leaving gaps in assessing generation fidelity.

To address these gaps, our evaluation framework combines multi-level automatic metrics with LLM-based rubric scoring to capture lexical, syntactic, and semantic accuracy together with human assessment to analyze alignment, enabling a more complete understanding of generation quality.

3 Evaluation Workflow

As shown in Figure 2, our workflow has three stages: (1) *Contextual Equation Corpus Creation* (§3.1), where equations are preserved within original narrative contexts from research papers.

(2) *Context-to-Equation Generation* (§3.2), where LLMs generate equations and descriptions based on the corpus. (3) *Multi-Level Equation Evaluation* (§3.3), assessing outputs via automatic metrics, LLM-based rubrics, and human evaluation.

3.1 Contextual Equation Corpus Creation

Data Source. The dataset was constructed to comprehensively cover recent research in AI domain, with a focus on papers containing significant mathematical or formal content. Papers were sourced from leading AI conferences archived in the ACL Anthology and OpenReview, including representative venues such as ACL and ICLR. We collected papers published in the past two years and applied selection criteria to ensure high-quality content, including methodological significance and the presence of multiple equations, ultimately including 170 papers in the dataset.

Dataset Construction. We extracted the first four pages of short papers and the first seven pages of other papers with equations.

We converted PDFs to Markdown using the META Nougat tool (Blecher et al., 2024), which preserves LaTeX equations and tables, removed files with failed conversions or with fewer than 2 or more than 15 equations. We merged multiple expressions appearing in a single block into one entry to preserve the grouping of related equations. The data schema is provided (see Appendix C).

Formally, the dataset is $\mathcal{D} = \{p_1, \dots, p_n\}$, where n denotes the total number of selected papers. Each paper p_i is extracted as a list of n_i context-equation-description triples: $p_i = [(c_1, e_1, d_1), \dots, (c_{n_i}, e_{n_i}, d_{n_i})]$, where c_j is the natural language context of the j -th equation, e_j is the corresponding LaTeX equation, and d_j is a concise description of variables in e_j .

Venue	#Paper	Avg #Eq	2–5 Eqs	6–10 Eqs	11+ Eqs
ACL	21	5.3	12	6	3
NAACL	28	5.1	17	9	2
NeurIPS	32	7.8	11	10	11
ICLR	46	7.8	11	25	10
ICML	43	8.1	14	18	11
Total	170	6.1	65	68	37

Table 1: Equation statistics by venues.

Dataset Statistics. After preprocessing and filtering the collected papers, we analyzed the distribution of equations to understand typical usage patterns. We analyzed 170 papers that each con-

Algorithm 1 Equation Generation Workflow

Require: Paper set $P = [p_1, \dots, p_n]$, n_i is max number of equations in i -th paper.
Initialize state:
1: $S.\text{equations} \leftarrow []$ $\triangleright$ Store generated equations and descriptions
2: $S.\text{current_paper} \leftarrow 0$
3: $S.\text{current_step} \leftarrow 0$
4: $S.\text{finished} \leftarrow \text{False}$
5: **for** $p_i \in P$ **do**
6: $M \leftarrow \text{EXTRACT}(p_i)$ $\triangleright$ Extract core methodology section
7: $\{c_1, \dots, c_{n_i}\} \leftarrow \text{DETECT_CONTEXT}(M)$ $\triangleright$ Original context
8: $\{e_1, \dots, e_{n_i}\} \leftarrow \text{DETECT_EQUATION}(M)$ $\triangleright$ Original equations
9: $\{d_1, \dots, d_{n_i}\} \leftarrow \text{DETECT_DESCRIPTION}(M)$ $\triangleright$ Original descriptions
10: **for** $j = 1$ to n_i **do**
11: $\mathcal{I}_j \leftarrow \text{CONSTRUCT_INPUT}(c_{1:j}, e_{1:j-1}, d_{1:j-1})$
12: $(e_j^g, d_j^g) \leftarrow \text{LLM}(\mathcal{I}_j)$ $\triangleright$ Generate equation and description
13: $S.\text{equations.append}((e_j^g, d_j^g))$
14: $S.\text{current_step} \leftarrow S.\text{current_step} + 1$
15: **end for**
16: $S.\text{current_paper} \leftarrow S.\text{current_paper} + 1$ $\triangleright$ Move to next paper
17: **end for**
18: $S.\text{finished} \leftarrow \text{True}$
19: **return** $S.\text{equations}$

tained two or more equations, totaling 1,043 equations. As shown in Table 1, most papers have 2 to 10 equations, while 37 papers have 11 or more. By venue, ICML, ICLR, NeurIPS papers are more equation-heavy (avg. 7.8–8.1 eqs/paper) than ACL and NAACL (avg. 5 eqs/paper), reflecting different writing styles.

3.2 Explainable Equation Generation

We formulate a *context-to-equation generation* task where an LLM generates a mathematical equation-description pair conditioned on the paper’s narrative context and previously generated outputs. Formally, given a paper $p_i \in \mathcal{D}$, the model generates the j -th equation-description pair,

$$(e_j^g, d_j^g) = \text{LLM}(c_{1:j}, e_{1:j-1}, d_{1:j-1}), \forall j \in [1, n_i],$$

where $c_{1:j}$ denotes all context segments up to step, $e_{1:j-1}, d_{1:j-1}$ are the previous equations and descriptions of variables.

As shown in Algorithm 1, it is formulated as a document-level, sequential generation problem: at each step j , an LLM conditions on the current textual context together with all previously introduced equations and variable descriptions, and generates the next equation-description pair (e_j^g, d_j^g) . The input is embedded into a standardized prompt template that enforces the required output format, instructing the model to return only the LaTeX code for the equation and a concise variable description. The full prompt and an example generation process are provided in Appendix E.

3.3 Multi-Level Equation Evaluation

Our evaluation protocol focuses on three primary aspects: automatic metrics, rubric metrics, and hu-

man judgment alignment.

Automatic Metrics. We assess equation quality using five automatic metrics: (1) **TexBLEU** (Jung et al., 2025) evaluates n-gram similarity with GPT-2 embeddings to account for both token-level and structural overlap; (2) **ROUGE-L** (Lin, 2004) measures longest common subsequence alignment; (3) Levenshtein distance (**Levenshtein-D**) (Levenshtein, 1966; Li and Liu, 2007) captures character-level edit costs normalized by length; (4) Sequence similarity (**SeqSim**), also called sequence matcher ratio (Rao et al., 2018) quantifies contiguous matching subsequences; to capture structural accuracy; (5) **Tree edit distance (TED)** (Schwarz et al., 2017) compares symbolic parse trees and averages normalized distances across equations within an entry. The range of all metrics are from 0 to 1, where higher values indicate greater similarity, except TED where 0 indicates identical expressions and 1 indicates maximum difference.

Rubric Metrics. We define five-dimensional rubrics for LLM and human judges to evaluate both surface-level correctness and deeper semantic and contextual alignment: (1) **Validity**, examining whether the equation is syntactically and mathematically well-formed; (2) **Coverage**, measuring whether all essential components are included; (3) **Clarity**, evaluating the clarity of the generated accompanying description; (4) **Appropriateness**, evaluating the degree to which the generated equation and description appropriately match the scenario, intent, or constraints of the original problem statement; (5) **Equivalence**, assessing whether the generated equation conveys the same mathematical meaning as the ground-truth equation, allowing for algebraically equivalent transformations and variable renaming.

For each rubric metric, we provide detailed instructions in Appendix D and specific information to guide its evaluation in Table 2.

Dimension	Context	GT_Eq*	GT_Des*	Gen_Eq	Gen_Des
Validity	✗	✗	✗	✓	✗
Coverage	✓	✗	✗	✓	✓
Clarity	✓	✓	✓	✓	✓
Appropriateness	✓	✗	✗	✓	✓
Equivalence	✓	✓	✓	✓	✓

Table 2: Information used in each LLM-as-Judge evaluation dimension. * indicates ground-truth; unmarked columns correspond to generated outputs.

Human Judgment Alignment. In addition to LLM-based evaluation, we conducted a human

Model	texBLEU (%)	Levenshtein (%)	SeqSim (%)	ROUGE-L (%)	Lexicality (%)	TED (%)	Syntax (%)
GPT-4.1	57.4	45.6	55.1	57.8	54.0	69.3	30.7
DeepSeek-R1	57.2	41.5	51.0	54.7	51.1	71.1	28.9
GPT-4o-mini	54.1	42.3	51.3	52.8	50.1	71.0	29.0
LLaMA3-70B	54.6	42.5	51.1	52.3	50.1	70.7	29.3
Qwen3-235B	59.8	44.1	53.1	55.1	53.0	69.9	30.1
Overall	56.6 $\pm$ 2.1	43.2 $\pm$ 1.5	52.3 $\pm$ 1.6	54.5 $\pm$ 2.0	51.7 $\pm$ 1.6	70.4 $\pm$ 0.7	29.6 $\pm$ 0.7

Table 3: Automatic evaluation results across lexical (TexBLEU, Levenshtein-D, SeqSim, ROUGE-L) and syntactic (1 - TED) metrics. Lexicality is the mean of the four lexical metrics, capturing token- and sequence-level agreement, while Syntax measures hierarchical structure recovery.

evaluation to examine alignment between LLMs and human judgments. A total of 100 examples from the evaluation dataset were independently assessed using the same five rubrics as the LLM-as-Judge. The evaluation was distributed across five human judges, and their assessments were aggregated to create a combined reference for comparison. Human judges were provided with the same rubrics available to LLMs. We assess the alignment by correlation coefficients: (1) **Pearson’s r** : Evaluates linear correlation between continuous variables. (2) **Spearman’s ρ** : Measures monotonic relationships between ranked variables. And (3) **Kendall’s τ** : Assesses ordinal associations between ranked variables.

4 Experimental Setup

4.1 Research Questions

This work is guided by the following research questions: **(RQ1)** How accurately can LLMs generate correct mathematical equation from unstructured real-world scientific literature? **(RQ2)** How does the number of equations in the original text influence the accuracy of the generation? **(RQ3)** how well do LLM evaluation align with human assessments in generation tasks?

4.2 Backbone Models

We evaluate equation generation performance across a diverse set of LLMs, covering both general-purpose and math-oriented architectures to enable comparative analysis. Specifically, we include the following models: (1) OpenAI GPT-4.1, (2) OpenAI GPT-4o-mini, (3) DeepSeek-R1, (4) LLaMA3.3-70B, (5) Qwen3-235B. OpenAI ² and DeepSeek ³ models are accessed via their native APIs, while LLaMA and Qwen models are queried through the Together AI platform ⁴.

²<https://platform.openai.com/docs/models>

³<https://api.deepseek.com/models>

⁴<https://www.together.ai/>

4.3 Implementation details

The proposed workflow is implemented by Python. During the equation generation stage, we use five representative LLM with their respective default decoding configurations. Specifically, Qwen3 is used with its default temperature of 0.7; GPT-4.1 and GPT-4o are used with temperature set to 1.0; and LLaMA-3 and DeepSeek-R1 are used with temperature set to 0.6. The input is formatted using a standardized prompt template (see Appendix E) to enforce consistent LaTeX and variable description outputs. All generated equations and descriptions are stored in structured JSON files indexed by paper and equation identifiers. To complement automatic metrics, we use GPT-4o-mini as an evaluator with a fixed temperature of 0.2. Each evaluation prompt provides the same contextual information used during generation, ensuring reliable and easily parseable LLM scoring. Detailed evaluation prompt templates are provided in Appendix F.

5 Main Results

5.1 Accuracy of Equation Generation (RQ1)

5.1.1 Evaluation with Automatic Metrics

Table 3 reports all models’ equation generation performance using automatic metrics. First, the models achieve moderate accuracy in recovering mathematical expressions from unstructured scientific text. The highest scores across four lexical similarity metrics remain below 60%, indicating persistent challenges in reliably generating mathematical equation from real-world scientific writing. Second, *Syntax* is substantially lower than *Lexicality* ($\approx 20\%$ higher error), reflecting the difficulty of correctly capturing the hierarchical structure of mathematical expressions. Third, standard deviations across models are small for all metrics, indicating little variation between models. These results partially answer (RQ1) by establishing quantitative upper bounds on current LLM accuracy. While models demonstrate basic competence, their

performance remains insufficient for highly precise scientific use, primarily due to syntactic errors.

5.1.2 Evaluation with Rubric Metrics.

Figure 3 show the LLM rubric evaluation results.

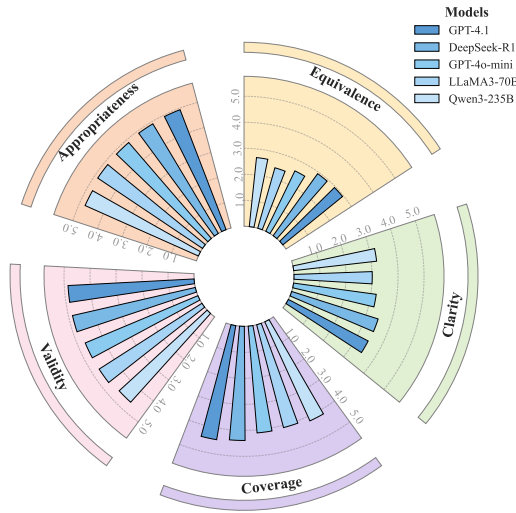


Figure 3: LLM-as-Judge average scores across metrics.

First, models generally produce well-formed equations that fit the input context, as reflected in high *Validity* and contextual *Appropriateness* scores around 4.7 to 4.9. However, *Equivalence* remains low across all models, ranging from approximately 2.4 to 3.0, and description clarity is moderate, around 3.0 to 3.5, indicating limitations in generating logically sound outputs.

Second, components *Coverage* scores are relatively high, approximately 4.1 to 4.5, suggesting that most outputs contain sufficient components to resemble plausible answers even when they are not fully correct.

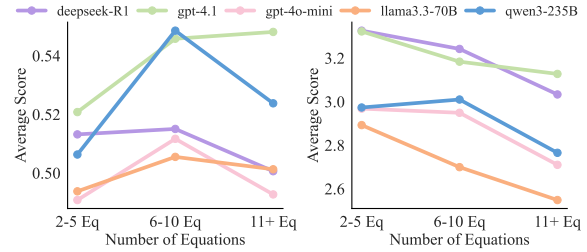
Third, scores of *Clarity* scores of the generated descriptions are relatively low, indicating that the descriptions do not clearly explain the variables in the generated equations or exhibit some inconsistencies with the original equations and descriptions.

Overall, these results show that while LLMs can generate contextually appropriate and syntactically correct formulas, their ability to faithfully reproduce the intended mathematical meaning is limited, highlighting challenges in accurately capturing the semantics of scientific equations and directly addressing (RQ1).

5.2 Effect of Number of Equations (RQ2)

To analyze the effect of the number of equations on generation accuracy, papers are grouped into

three categories based on equation count: 2–5, 6–10, and 11 or more. We examine the average performance using both automatic metrics and LLM-based rubric metrics (see Figure 4).



(a) Automatic metrics.

(b) LLM rubric metrics.

Figure 4: Effect of equation count on generation accuracy. Papers are grouped by equation numbers: 2–5, 6–10, and 11+.

The results of automatic metrics show a slight increase for papers with 6–10 equations, followed by a decrease for papers with more than 10 equations. This result suggests that a moderate number of equations may provide sufficient context for models to better match the form of equations, while a larger number of equations may increase challenges that reduce lexical and syntactic accuracy.

In contrast, LLM-based evaluations show a clear downward trend as the number of equations increases, indicating that larger numbers of equations make it more difficult for models to maintain semantic accuracy and reasoning quality.

These results indicate that the increase in equation numbers negatively impacts the performance of LLMs, particularly in capturing the intended meaning of mathematical expressions, and that automatic metrics may underestimate the impact of equation density on accuracy.

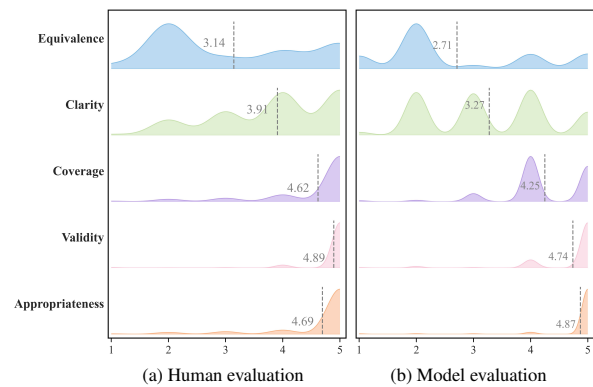


Figure 5: Comparison of the distribution of human-evaluated scores and model-evaluated scores.

5.3 Alignment with Human Judgments (RQ3)

Figure 5 shows that human judges consistently assign higher scores than LLM-as-Judge across all five evaluation metrics, including semantic accuracy, description clarity, component coverage, syntax validity, and context alignment.

Correlation analysis reveals that DeepSeek-R1 achieves the highest alignment with human judgments, with moderate Spearman and Kendall’s tau correlations for semantic accuracy (see Figure 6). Other models and dimensions exhibit generally weak correlations, with many near zero or negative, indicating inconsistent ranking behavior compared to humans. Cohen’s kappa values are uniformly low across all models, mostly around 0.2, reflecting moderate agreement in exact score assignment.

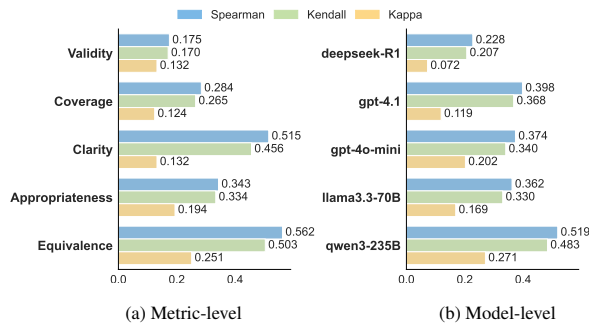


Figure 6: Alignment between LLM-based and human evaluations measured by correlation coefficients.

Overall, these results suggest that LLM-as-Judge provides limited alignment with human evaluation, showing partial consistency only in certain models and dimensions, and highlights the challenges of relying on automated judgments to fully replicate human assessment, directly answering (RQ3).

6 Additional Analysis

6.1 Performance Over Different Venues

Figure 7 shows evaluation scores across five venues, using both automatic metrics and LLM-as-Judge rubric-based scores.

First, the scores demonstrate minimal variation across venues: no conference consistently yields higher or lower values on any of the ten metrics. This result indicates that differences in writing style and equation density among these venues do not substantially affect the evaluation outcomes.

Second, *Syntax* and *Equivalence* consistently receive lower scores than other metrics in all venues, highlighting the difficulty of capturing mathematical structure and symbolic equivalence beyond

surface-level formula understanding and matching.

6.2 Evaluation with Explanations

Besides generating scores, our LLM-based evaluation framework produces natural-language explanations for each rating. The explanations provide insight into the specific reasons behind a low or high rating, highlighting differences in variable usage, equation structure, or semantic interpretation.

Context	Here, ops is a sequence of edit operations transforming G_1 into G_2 , with $w(op_i)$ representing the cost of the i -th operation. Normalization: The tree edit distances are normalized to account for the complexity of the code by considering the maximum number of nodes between the two trees. A ramp function is added to avoid extreme values.
Reference	$TSED = \max\{1 - \frac{\delta}{\maxNodes(G_1, G_2)}, 0\}$
Generation	$TSED(G_1, G_2) = \min\left(\frac{\Delta(G_1, G_2)}{\max(G_1 , G_2 , 1)}, 1\right)$
Score	1
Explanation	The generated equation fundamentally alters the relationship expressed in the ground truth by using a minimum instead of a maximum, leading to a significant misunderstanding of the intended normalization process.

Table 4: Example of evaluation outcome with explanation.

For instance, in Table 4, the low score 1 on equivalence is caused by an incorrect normalization process: The generated equation uses “min” instead of “max”, altering the semantics of the reference formulation. By offering interpretable feedback alongside numerical scores, this reasoning-enhanced evaluation not only increases transparency but also facilitates qualitative analysis of LLM outputs, allowing researchers to better understand where and why generated equations deviate from the intended meaning. Overall, integrating reasoning explanations improves the utility of LLM-as-Judge for evaluating equation generation, making the assessment more informative and actionable.

7 Discussion and Implication

Complementary Evaluation Fidelity. We combine automatic metrics, LLM-based rubrics, and human judgments to evaluate equation accuracy, explainability, and alignment. Results show that single-metric evaluation is biased, with mathematical unit identification and symbolic semantics remaining key bottlenecks. Models often misidentify core variables, causing semantic errors despite well-formed expressions, indicating a tendency to favor surface plausibility over semantic accuracy.

Model Capacity vs. Equation Density. Our results provide nuanced insights into how LLMs pro-

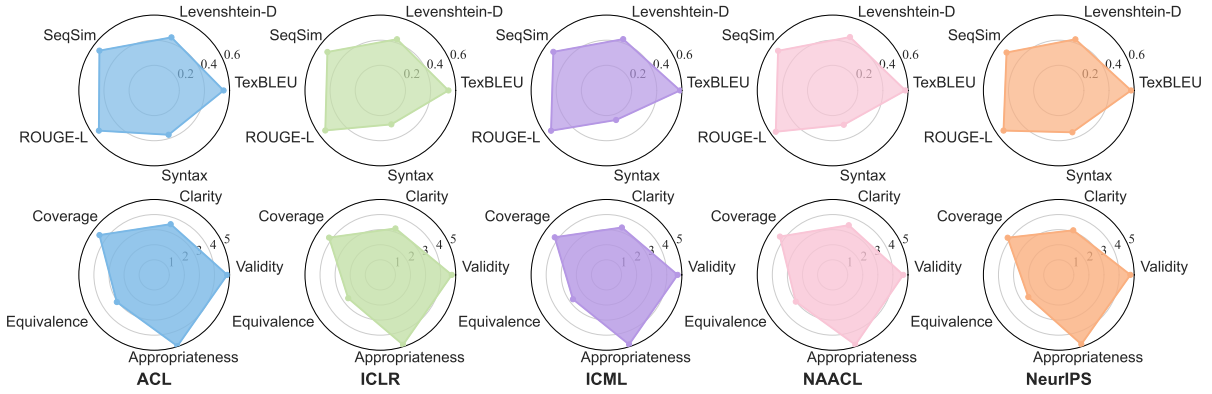


Figure 7: Scores for different venues.

cess structured mathematical content embedded in unstructured text, revealing a clear interaction between model capacity and equation density. We observe a non-monotonic effect of equation density on generation performance. Generation performance improves slightly for papers with a moderate number of equations (6–10), indicating helpful contextual cues, but declines as density increases, especially for semantic accuracy, suggesting limitations in maintaining long-range symbolic dependencies.

Minimal Venue-Specific Writing Style Effects.

Although the number of equations varies by venue – ICML, ICLR, and NeurIPS are more equation-heavy (around eight equations per paper) than ACL and NAACL (around five) – model performance remains largely stable across venues (§5.2). Evaluation scores are also consistent across venues (§5.3). These results suggest that the observed limitations arise from inherent model capabilities rather than writing style or venue-specific conventions. We caution against treating venue characteristics as proxies for scientific creativity or difficulty.

Implications. These findings indicate the need for evaluation frameworks and modeling approaches that explicitly account for symbolic structure, inter-equation dependencies, and semantic equivalence, beyond surface-level textual similarity. **SciText2Eq** provides a general and extensible workflow for both modeling and evaluation in equation generation. Built from recent AI papers, it is not domain-specific, and its methods for contextual equation corpus construction, context-to-equation generation, and multi-level evaluation can apply to other formal scientific domains such as physics, biology, economics, and engineering.

Future Work. We plan to extend the workflow to enable cross-paper reasoning by incorporating a retriever that surfaces related equations together with their original contexts from other papers. Another promising direction is to apply reinforcement learning to optimize equation generation with respect to both syntactic structure and semantic correctness. Finally, future work may improve evaluator alignment by combining rubric-based LLM judgments with lightweight human-in-the-loop calibration, enabling more reliable evaluation of scientific equation generation.

8 Conclusion

In this work, we investigated the ability of LLMs to generate mathematical equations from real scientific texts. We introduce an explainable equation generation framework evaluated on diverse open- and closed-source LLMs, along with a unified evaluation protocol combining automatic metrics, LLM-based rubrics, and human judgments to measure accuracy, explainability, and human-LLM alignment. Our analyses reveal that LLMs can produce syntactically valid and contextually appropriate equations. Meanwhile, they often struggle with semantic accuracy in scientific documents with high equation density. By framing equation generation as a proxy for creative reasoning, we highlight how accurate extraction and reconstruction of mathematical expressions is essential for identifying, combining, and building upon conceptual relationships across studies. Overall, our work highlights the need for improved models and evaluation strategies to better capture the semantic and structural differences of scientific equations, paving the way toward AI-assisted scientific discovery.

Limitations

Our work reveals several limitations of current LLMs in generating mathematical equations from scientific texts. Despite moderate performance on lexical- and syntactic-based metrics, models struggle with semantic accuracy, particularly when papers contain a higher density of equations. This gap underscores the difficulty of accurately interpreting and reconstructing mathematical content in realistic scientific contexts, beyond the simplified settings of existing benchmarks. These limitations have direct implications for scientific creativity. Generating equations from unstructured text can be viewed as a proxy for creative scientific reasoning, requiring the ability to identify, interpret, and recombine conceptual relationships across studies. When LLMs fail to fully capture the intended meaning of equations or miss subtle dependencies, their outputs may misrepresent the underlying concepts, potentially constraining the ability of researchers to synthesize knowledge efficiently and explore novel ideas. Thus, improving semantic understanding and reasoning in equation generation is essential not only for model performance but also for supporting human scientific creativity and accelerating the research process.

Ethical Consideration

This work focuses on evaluating and improving automated scientific equation generation and does not involve human subjects, personal data, or sensitive information, and we see no potential risks. All data are sourced from publicly available research papers and used solely for research purposes. While LLMs can generate mathematically plausible equations, they may also produce incorrect or misleading formulations; we therefore emphasize explainability and human-aligned evaluation to mitigate risks of misuse. Our evaluation framework is intended to support, not replace, expert judgment, and we encourage responsible use of these tools in scientific workflows where final validation remains with human researchers.

References

- Aida Amini, Saadia Gabriel, Shanchuan Lin, Rik Koncel-Kedziorski, Yejin Choi, and Hannaneh Hajishirzi. 2019. *Mathqa: Towards interpretable math word problem solving with operation-based formalisms*. In *Proceedings of the 2019 conference of the North American chapter of the association*
- for computational linguistics: Human language technologies (NAACL-HLT), pages 2357–2367.
- Sander Bais. 2025. *The equations: icons of knowledge*. Routledge.
- Lukas Blecher, Guillem Cucurull, Thomas Scialom, and Robert Stojnic. 2024. *Nougat: Neural optical understanding for academic documents*. In *The Twelfth International Conference on Learning Representations (ICLR)*.
- Ting-Rui Chiang and Yun-Nung Chen. 2019. *Semantically-aligned equation generation for solving and reasoning math word problems*. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: Human language technologies (NAACL-HLT)*, pages 2656–2668.
- Junlan Dong and Jinghui Zhong. 2025. *Recent advances in symbolic regression*. *ACM Computing Surveys*, 57(11):1–37.
- Mengge Du, Yuntian Chen, Zhongzheng Wang, Longfeng Nie, and Dongxiao Zhang. 2024. *Large language models for automatic equation discovery of nonlinear dynamics*. *Physics of Fluids*, 36(9).
- Kyudan Jung, Nam-Joon Kim, Hyun Gon Ryu, Sieun Hyeon, Seung-Jun Lee, and Hyuk-Jae Lee. 2025. *Texbleu: Automatic metric for evaluating latex format*. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5.
- John R Koza. 1994. *Genetic programming as a means for programming computers by natural selection*. *Statistics and computing*, 4(2):87–112.
- Vladimir I Levenshtein. 1966. *Binary codes capable of correcting deletions, insertions, and reversals*. *Soviet Physics Doklady*, 10(8):707–710.
- Yujian Li and Bo Liu. 2007. *A normalized levenshtein distance metric*. *IEEE transactions on pattern analysis and machine intelligence*, 29(6):1091–1095.
- Zhenwen Liang, Wenhao Yu, Tanmay Rajpurohit, Peter Clark, Xiangliang Zhang, and Ashwin Kalyan. 2023. *Let gpt be a math tutor: Teaching math word problem solvers with customized exercise generation*. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 14384–14396.
- Chin-Yew Lin. 2004. *Rouge: A package for automatic evaluation of summaries*. In *Text summarization branches out*, pages 74–81.
- Jordan Meadows, Marco Valentino, and André Freitas. 2025. *Controlling equational reasoning in large language models with prompt interventions*. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, volume 39, pages 24858–24866.

- Matteo Merler, Katsiaryna Haitsiukevich, Nicola Dainese, and Pekka Marttinen. 2024. [In-context symbolic regression: Leveraging large language models for function discovery](#). *arXiv preprint arXiv:2404.19094*.
- Shuai Peng, Ke Yuan, Liangcai Gao, and Zhi Tang. 2021. [Mathbert: A pre-trained model for mathematical formula understanding](#). *arXiv preprint arXiv:2105.00377*.
- Brenden K Petersen, Mikel Landajuela, T Nathan Mundhenk, Claudio P Santiago, Soo K Kim, and Joanne T Kim. 2021. [Deep symbolic regression: Recovering mathematical expressions from data via risk-seeking policy gradients](#). In *International Conference on Learning Representations (ICLR)*.
- G. Appa Rao, G. Srinivas, K. Venkata Rao, and P. V. G. D. Prasad Reddy. 2018. [Characteristic mining of mathematical formulas from document: A comparative study on sequence matcher and levenshtein distance procedure](#). *International Journal of Computer Sciences and Engineering*, 6(4):400–404.
- Alexander Scarlatos and Andrew Lan. 2023. [Tree-based representation and generation of natural and mathematical language](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 3714–3730.
- Stefan Schwarz, Mateusz Pawlik, and Nikolaus Augsten. 2017. [A new perspective on the tree edit distance](#). In *International Conference on Similarity Search and Applications*, pages 156–170. Springer.
- Parshin Shojaei, Kazem Meidani, Shashank Gupta, Amir Barati Farimani, and Chandan K Reddy. 2025a. [Llm-sr: Scientific equation discovery via programming with large language models](#). In *The Twelfth International Conference on Learning Representations (ICLR)*.
- Parshin Shojaei, Ngoc-Hieu Nguyen, Kazem Meidani, Amir Barati Farimani, Khoa D Doan, and Chandan K Reddy. 2025b. [Llm-srbench: A new benchmark for scientific equation discovery with large language models](#). In *Proceedings of the Twelfth International Conference on Machine Learning (ICML)*.
- Zhilong Song, Qionghua Zhou, Chunjin Ren, Chongyi Ling, Minggang Ju, and Jinlan Wang. 2025. [Llm-feynman: leveraging large language models for universal scientific formula and theory discovery](#). *arXiv preprint arXiv:2503.06512*.
- Sumedh Vemuganti, Ayu Seiya, and Nickvash Kani. 2025. [Advancing math formula search using diverse structural and symbolic representations](#). In *Advances in Information Retrieval: 47th European Conference on Information Retrieval (ECIR)*, pages 116–131. Springer.
- Runxiang Wang, Boxiao Wang, Kai Li, Yifan Zhang, and Jian Cheng. 2025. [Drshr: Llm based scientific equation discovery with dual reasoning from data and experience](#). *arXiv preprint arXiv:2506.04282*.
- Yan Wang, Xiaojiang Liu, and Shuming Shi. 2017. [Deep neural solver for math word problems](#). In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 845–854.
- Zichao Wang, Mengxue Zhang, Richard G Baraniuk, and Andrew S Lan. 2021. [Scientific formula retrieval via tree embeddings](#). In *2021 IEEE International Conference on Big Data (Big Data)*, pages 1493–1503. IEEE.
- Michihiro Yasunaga and John D Lafferty. 2019. [Topicq: A joint topic and mathematical equation model for scientific texts](#). In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, volume 33, pages 7394–7401.
- Zhouliang Yu, Ruotian Peng, Keyi Ding, Yizhe Li, Zhongyuan Peng, Minghao Liu, Yifan Zhang, Zheng Yuan, Huajian Xin, Wenhao Huang, Yandong Wen, Ge Zhang, and Weiyang Liu. 2025. [Formal-math: Benchmarking formal mathematical reasoning of large language models](#). *arXiv preprint arXiv:2505.02735*.
- Wenqi Zhang, Yongliang Shen, Qingpeng Nong, Zeqi Tan, Yanna Ma, and Weiming Lu. 2023. [An expression tree decoding strategy for mathematical equation generation](#). *arXiv preprint arXiv:2310.09619*.
- Mingle Zhou, Ming Cai, Gang Li, and Min Li. 2022. [An end-to-end formula recognition method integrated attention mechanism](#). *Mathematics*, 11(1):177.
- Mingyu Zong and Bhaskar Krishnamachari. 2023. [Solving math word problems concerning systems of equations with gpt-3](#). In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, volume 37, pages 15972–15979.

A Most Related Works

Table 5 summarizes the most relevant prior work in equation discovery and equation generation. Our proposed workflow, **SciText2Eq**, is also listed for comparison, highlighting its focus on context-aware generation and multi-level evaluation in the AI domain.

B Usage Disclosure

AI tools were used only in a limited capacity to assist with language editing. Specifically, we use GPT-5.2 to improve the clarity and readability of the manuscript. All scientific contributions, including research design, data collection, analysis, results, and conclusions, have been independently conducted and verified by the authors.

Method	Domain	#Sample	Input	Output	Core Metric
DrSR	Physics, Chemistry, Biology, Materials Science, etc.	4000 equations	• Variable pairs • Scientific priors	• Symbolic equation	• ACC_{τ} /NMSE
LLM4ED	Physics	22 tasks	• Variable pairs • Symbol library	• Symbolic equation	• NRMSE
LLM-Feynman	Physics, Materials Science	120 equations + 4 tasks	• Variable pairs • Scientific priors	• Symbolic equations • Complexity • Interpretability score • Symbolic equations	• MAE • Interpretability Score*
LLM-SRBench	Physics, Chemistry, Biology, Materials Science	239 equations	• Variable pairs • Problem description • Scientific priors	• Symbolic equations	• ACC_{τ} /NMSE • Symbolic Accuracy*
LLM-SR	Physics, Micro-biology/Biology, Materials Science, etc.	-	• Variable pairs • Problem description • Scientific priors	• Equation skeleton • Optimized parameters	• NMSE
ICSR	General symbolic - regression	-	• Variable pairs	• Symbolic equation • Optimized parameters	• R^2 /Complexity
TopicEq	physics, computer science, astronomy, etc	400k equations	• Equations • Surrounding text	• Symbolic equation • Inferred topics	• Topic Coherence • Perplexity • Syntax Error Rate
SciText2Eq	AI	1043 equations	• Context • Variable description • Prior equations	• Symbolic equation • Generated description	• Topic Coherence • Perplexity • Syntax Error Rate

Table 5: Comparison of prior symbolic regression / scientific equation generation works. Metrics marked with * employ LLM as a judge for evaluation.

C Dataset Structure

We organize our dataset in a structured JSON format, capturing the hierarchical relationships between papers, equations, and their contextual information. Each entry represents a single paper, identified by a unique ID derived from its filename.

Our extraction process is **section-aware**: we draw context from the methods section to maintain semantic coherence across equations, and we systematically clean equations to remove extraneous or presentation-only LaTeX formatting. In addition, variable definitions are explicitly captured to preserve explanatory detail and improve interpretability.

To represent mathematical content in a structured manner, we group related equations into **equation groups**. Each group is associated with a unique equation ID and includes the preceding contextual text, variable descriptions (typically sentences starting with "where"), and an array of LaTeX-formatted equations. This design enables faithful representation of cases where multiple equations share a common context, such as systems of equations or step-by-step derivations. Finally, all extracted equations and their associated metadata are serialized into a structured JSON file, with an example entry shown below:

```
{
  "id": "2024.acl-short.15",
  "title": "Code-Switching Can be
    Better Aligners",
  "equations": [
    {
      "equation id": "1",
      "context": "Describes the joint
        SLU model for intent ...",
      "description": "where  $f(\cdot)$  is the
        joint model ...",
      "EQ_latex": [" $(o^{\{I\}}, o^{\{S\}}) =$ 
         $f(x)$ "]
    },
    {
      "equation id": "2",
      "context": "Introduces optimal
        transport (OT) for aligning
        ...",
      "description": "where  $\text{norm}(\cdot)$ 
        denotes row normalization
        ...",
      "EQ_latex": [" $\hat{Q}_{[i,j]} =$ 
         $\text{norm}(Q_{[i,j]})$ "]
    }
  ]
}
```

D Rubric Metric details

As shown in Table 6, we evaluate generated equations and descriptions using five rubric metrics. Each metric is rated on a 1–5 scale, with 5 being the best.

Metric, Description and Scores
Validity Measures if equations are well-formed LaTeX, independent of correctness. • 5: Fully valid; no syntax issues. • 4: Minor syntax issues; easily correctable. • 3: Noticeable formatting issues; still parseable. • 2: Multiple syntax errors hindering understanding. • 1: Completely ill-formed; not parseable.
Coverage Checks if equations and descriptions provide a full solution considering intermediate steps. • 5: All necessary terms, variables, and constraints present. • 4: Minor omissions; still practically usable. • 3: Noticeable omissions; partially interpretable. • 2: Several important components missing or ambiguous. • 1: Incomplete or disconnected from solution.
Clarity Evaluates logical clarity and correctness of inferred relationships and generate description. • 5: Clear and fully logical inferred reasoning and description. • 4: Generally logical; minor ambiguities. • 3: Partially clear; noticeable gaps or ambiguity. • 2: Significant inconsistencies; barely understandable. • 1: No coherent reasoning; confusing or nonsensical.
Appropriateness Assesses whether equations and descriptions match the scenario, intent, and constraints. • 5: Perfectly matches and clearly addresses context. • 4: Strong alignment; minor ambiguities. • 3: Partial relevance; some generic or inferred parts. • 2: Loosely related; insufficient follow-through. • 1: Completely irrelevant or hallucinated content.
Equivalence Measures whether the generated equation expresses the same mathematical relationships as ground truth. • 5: Exact same meaning; only trivial variations (algebraic rearrangement, variable renaming). • 4: Near-match; minor semantic deviations, intent clearly preserved. • 3: Core meaning largely correct but includes secondary inaccuracies. • 2: Partial overlap; significant misunderstanding or contradiction. • 1: Completely unrelated; wrong understanding.

Table 6: Rubric metrics for LLMs and human evaluation. Each metric is listed with its description and scoring criteria.

E Generation Prompt Templates

System Prompt

You are a scientific writing assistant trained to generate LaTeX equations for research papers. For each equation (Equation n), you will receive a combined context which includes:

- The surrounding natural language context for each equation up to Equation n .
- Any prior equations (Equation 1 to Equation $n-1$) along with their LaTeX representations, if $n > 1$.
- Descriptions for prior equations (Equation 1 to Equation $n-1$), if available.

Your task:

- (1) Understand the context and flow of the document.
- (2) Generate the LaTeX code for the next equation (Equation n) based on the combined context. Respond with the LaTeX code and the description only. Wrap LaTeX in `<latex>...</latex>` tags and wrap a one-sentence variable description in `<description>...</description>`. Do not include any explanation or commentary.

User Prompt

3 Preliminaries

Traditional works rely on text frequencies to define whether an instance is long-tail or not; thus, low-frequency texts tend to be classified into long-tail classes. For LLMs, computing text frequencies of previously unknown user queries is non-trivial. Following prior work (Aimar et al., 2023; Zhong et al., 2021; Xu et al., 2021), *Expected Calibration Error (ECE)* is used to quantify “long-tailness”. ECE measures how well predicted probabilities align with observed accuracies (Guo et al., 2017). Formally, ECE is defined as:

Example Output:

- `<latex> ECE = \sum_{i=1}^B \frac{n_{b_i}}{N} |acc(b_i) - conf(b_i)| </latex>`
- `<description>` where i denotes the i -th bin, N is the total number of instances, $acc(b_i)$ and $conf(b_i)$ are the accuracy and confidence of bin b_i , and n_{b_i} is the number of instances in bin b_i

Table 7: System and user prompt template for equation generation.

F Evaluation Prompt Templates

We adopt an LLM-as-a-judge framework to evaluate generated equations and their accompanying descriptions. Table 8 presents the evaluation prompt used across all rubric metrics. All evaluation prompts share the same overall structure and output format, and differ only in the evaluation focus (see Appendix D) and required input information (see Table 2).

System Prompt

You are a mathematical evaluation assistant trained to score generated equations and variable descriptions. For each evaluation dimension, you will receive:

- The problem context.
- The ground truth equations and their descriptions.
- The generated equations and their descriptions.

Your task:

- (1) Provide a score from 1 to 5 according to the specific evaluation dimension.
- (2) Provide a brief explanation. Respond **only** in the following format:
 - Score: <1-5>
 - Explanation: <brief justification>

User Prompt Template

Evaluation Metric: Equivalence

- **Context:** {context}
- **Ground Truth Equation(s):** {eq_gt}
- **Generated Equation(s):** {eq_gen}
- **Ground Truth Description(s):** {description_gt}
- **Generated Description(s):** {description_gen}

Task: Evaluate whether the generated equation expresses the same mathematical relationships as the ground truth, allowing for equivalent rearrangements or variable renaming.

Example Output:

- Score: 5
 - Explanation: The generated equation exactly matches the ground truth in mathematical meaning, with only trivial rearrangements.
-

Table 8: Evaluation prompt with Equivalence metric as an example.